

MODULE 1

POWER SUPPLIES

DC Power Supply: Most of the electronic devices and circuits are operated by DC power supplies. It consists of four main stages explained below.

Step down transformer: It is a device that has two coil windings: primary and secondary used to convert a high AC voltage (230V/ 50Hz) to a required low AC voltage.

Rectifier: It is a device that has one or more diodes, converts secondary AC voltage to pulsating DC.

Smoothing Filter: It is a circuit used to remove fluctuations (ripple or ac) present in rectifier output. Example: Capacitor filters, LC filters, π - filters, etc..

Voltage Regulator: Voltage regulator is a circuit which provides constant DC output voltage irrespective of changes in load current or changes in input voltage.

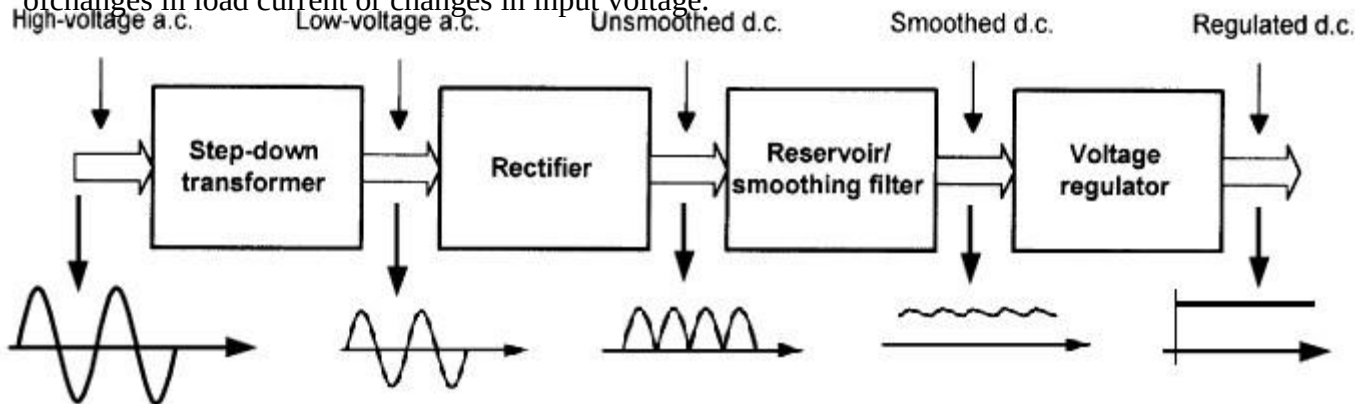


Fig.1. Block diagram of a DC power supply

Fig. 2, shows important electronic components that are used in the block diagram in fig. 1. Step-down transformer is made of iron core, feeds a rectifier. Rectifier output is applied to a high value capacitor to minimize ripples. Capacitor filter charges as the rectifier output voltage increases until its peak value. When the voltage value reduces, it discharges gradually through the regulator. Finally, a series transistor regulator and zener diode provides a constant output DC voltage.

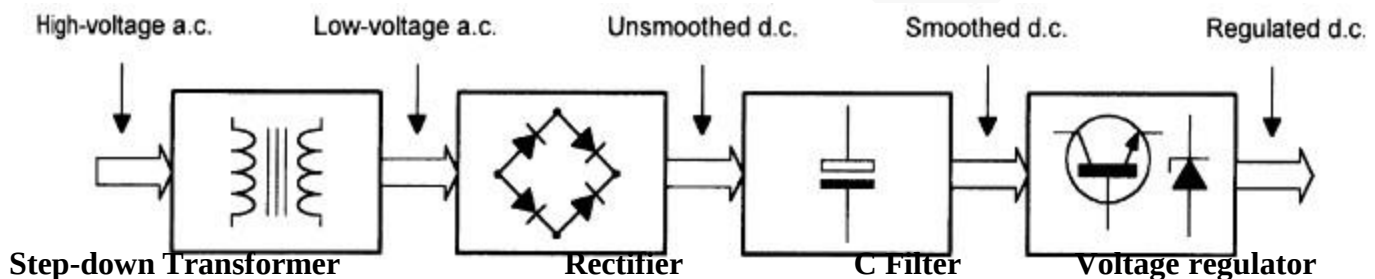


Fig.2. Block diagram of a DC power supply showing principal components used in each stage

Rectifiers

Semiconductor diodes are commonly used as rectifiers. It converts AC voltage into rippled DC voltage. There are two types: Half-wave and full wave rectifiers. Fig. 3 shows *half-wave* rectifier that allows one half of an AC waveform to pass through to the load. AC voltage (240V r.m.s) is applied to the primary of step-down transformer (T1). The secondary of T1, reduces to 12V r.m.s. (Taking turns ratio: 240: 12 = 20:1). Diode D1 will allow current only in positive half cycle being forward biased and operates as a closed switch, see fig. 3(b). For negative half cycle, current will not allow passing through D1, because it is reverse biased and act like an open switch, see fig. 3(c).

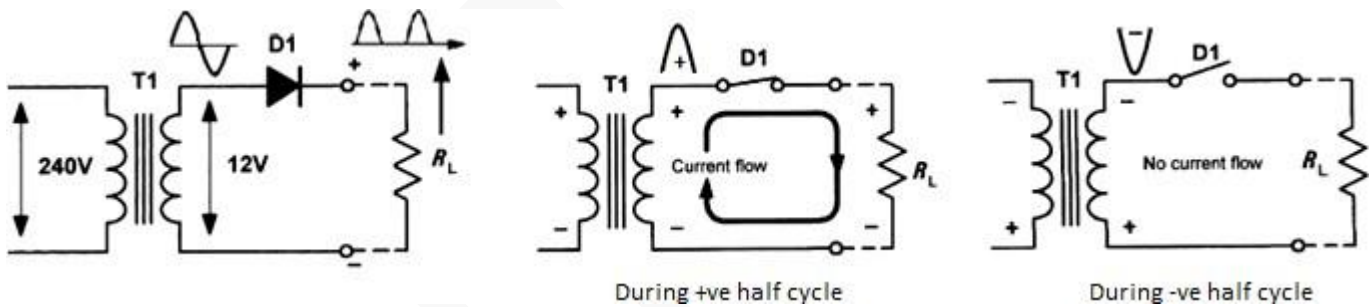


Fig.3. (a) Half wave rectifier circuit (b) For + ve half cycle (closed switch) (c) For - ve half cycle (open switch)

The switching action of D1 results in pulsating output voltage available at load resistor (R_L). During positive half cycle, silicon diode will drop 0.6V to 0.7V as forward threshold voltage. During negative half cycle, D1 is reverse biased, hence secondary of T1 peak voltage will be dropped across it.

Analysis During +ve half cycle:

Secondary of T1 = 12V r.m.s
 voltage Peak voltage across
 secondary windings: $V_{\text{peak}} = 1.414 \times V_{\text{rms}}$
 $= 1.414 \times 12 = 16.968\text{V} \sim 17\text{V}$
 Silicon diode drop voltage =
 0.7V Actual output voltage
 across load R_L
 $= 17 - 0.7 = 16.3\text{V}$

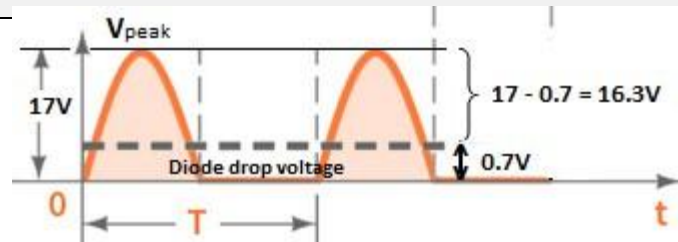
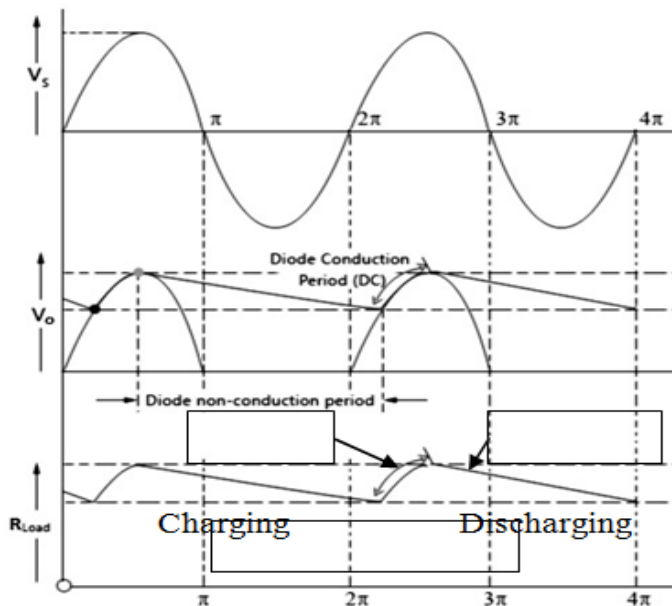


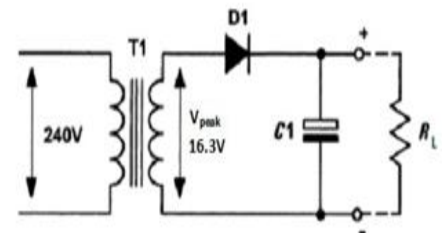
Fig.4 Illustration of actual output voltage across load R_L

Smoothing (Reservoir) circuits

Smoothing circuit is a capacitor filter C_1 connected in parallel to the load R_L as shown in the fig.5. It is used to remove fluctuations (ripple or ac) present in rectifier output. When 240V AC voltage is applied to primary of T1, its secondary reduces to 12V r.m.s value and peak value is 16.3V. During +ve half cycle of secondary voltage, diode is forward biased, C_1 charges as the rectifier output voltage increases to its peak value (16.3V). When the rectifier voltage starts to decrease, C_1 discharges slowly through the load R_L , until the next +ve half cycle is met.



Output voltage across R_L



Charging Time of C1 to the peak value = $R_{\text{series}} \times C1$
 $R_{\text{series}} = R_{\text{secondary winding}} + R_{\text{diode}} + R_{\text{wiring}}$
 and connections Hence C1 charges quickly as soon as diode conducts.

Discharging Time of C1 = $R_L \times C1$
 Practically, R_L is very much larger than R_{series}
 Hence C1 discharges slowly through R_L .

Capacitor as reservoir: C1 stores charge during +ve half cycle of secondary V_{peak} and releases it during -ve half cycle, maintaining reasonably constant output voltage across R_L . This causes to a small DC ripples at the output. The DC ripples can be drastically reduced by choosing a larger C1 value in place of smaller value.

Improved ripple filters In filters the value of the capacitor plays an important role in determining the output ripples and the average DC level. If the capacitor value is high, the amount of charge it can store will be high and the amount it discharges will be less. Thus the ripples will be less and the average dc level will be high.

Limitations of C filter

If the capacitor value is increased to a very high value, the amount of current required to charge the capacitor will be high. So, diodes are subjected to high surge currents. Thus, there is a limit in increasing the capacitor value in half-wave rectifiers.

Refinement of C filter (RC filter)

Additional components $R1$ and $C2$ are connected as shown in the fig.6. $C1$ and $C2$ offer low reactance to AC components of ripple. In effect $R1$ and $C2$ act like a voltage divider and amount of ripple is reduced. But certain amount of DC voltage will drop across $R1$. The value of $C2$ is selected in such way that it exhibits negligible reactance at low frequencies

(50Hz – 100Hz).

Amount of ripple reduction is determined by

$$\frac{X_C}{\sqrt{(R_1^2 + X_C^2)}}$$

Where, X_C = reactance of $C2$.

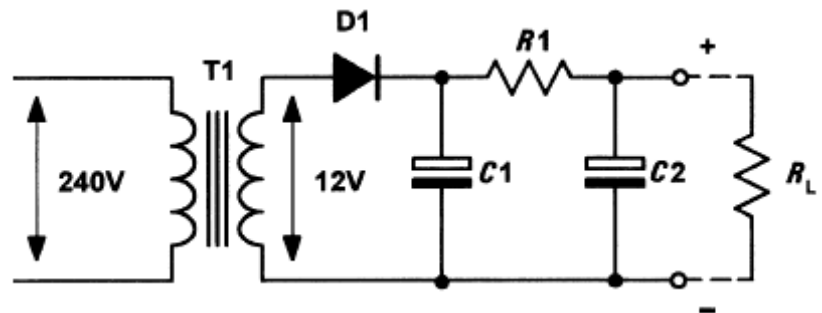


Fig.6. Half wave rectifier with $R1$ and $C2$

L-C Smoothing Filter

From the fig.7, at the ripple frequency, $C1$ exhibits low value of capacitive reactance. Hence it bypasses most of AC components of ripples. $L1$ exhibits high value of inductive reactance, therefore it allows most of DC components. Further, $C1$ bypasses remaining AC components offering low value of capacitive reactance. Thus the combined effect of L C greatly reduces the ripples.

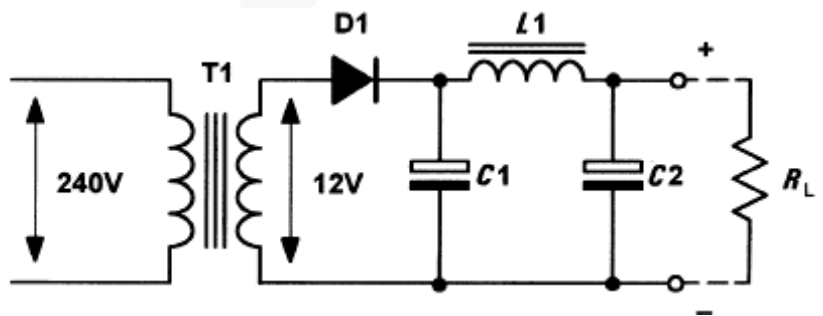


Fig.7. Half wave rectifier with $R1$ and LC filter

Advantage: Half wave rectifier is cheap, simple and easy to construct.

Disadvantage:

1. Ripple factor is high at the output.
2. Rectification efficiency is quite low, that means, power is delivered only during one half cycle of the input alternating voltage.
3. Transformer utilization factor is low.

Full-wave rectifiers

Full-wave rectifier – there are two types:

- *Bi-Phase or Center Tapped full wave rectifier* - uses two diodes and center tapped power transformer.
- *Bridge full wave rectifier* - uses four diodes and ordinary power transformer.

Bi-phase Rectifier

The AC mains (240V) is applied to the primary of T1 which has two identical secondary windings each providing 12V r.m.s, as shown in the fig.8.

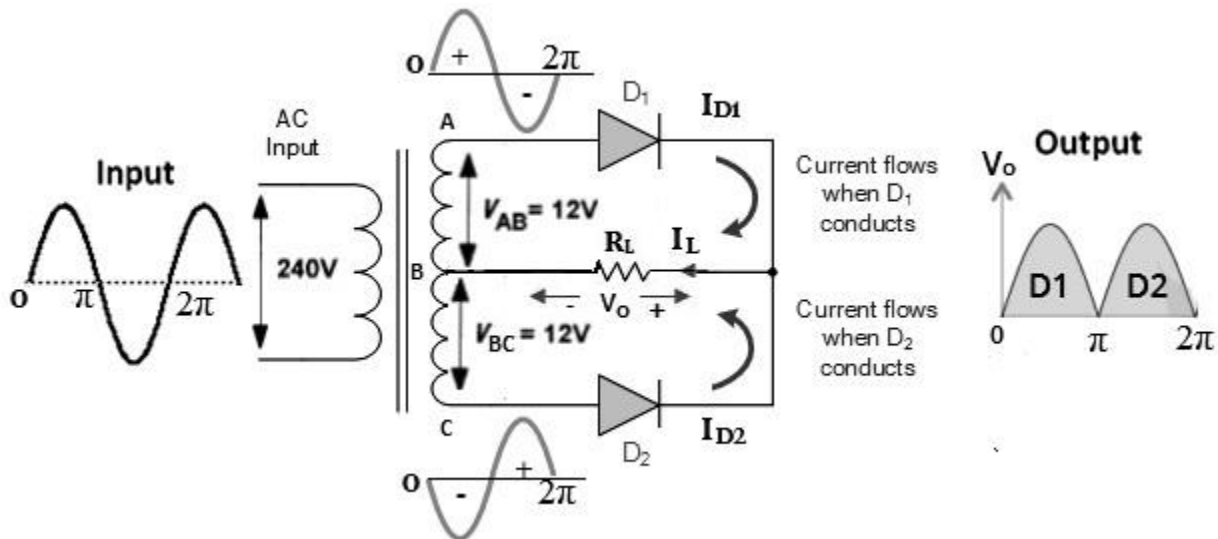
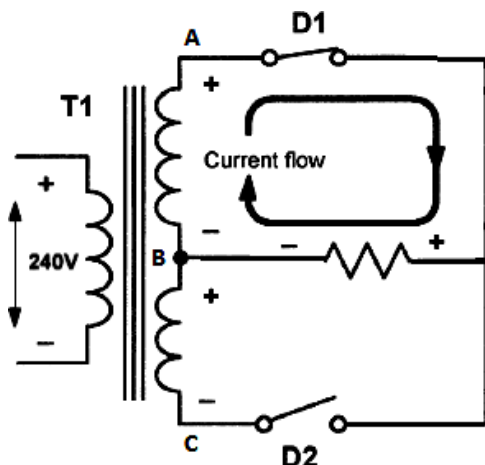


Fig.8. Bi-phase rectifier circuit

On +ve half cycles, point A will be +ve with respect to point B. similarly, point B will be +ve with respect to point C.

D1 will forward bias, acts like a closed switch hence conducts. While D2 will reverse bias, acts like an open switch hence do not conduct. It is as shown in the fig. 9(a).

Thus, D1 alone conducts on +ve half cycles.



On -ve half cycles, point C will be +ve with respect to point B. similarly, point B will be +ve with respect to point A.

D2 will forward bias, acts like a closed switch hence conducts. While, D1 will reverse bias, acts like an open switch hence do not conduct. It is as shown in the fig. 9(b).

Thus, D2 alone conducts on -ve half cycles.

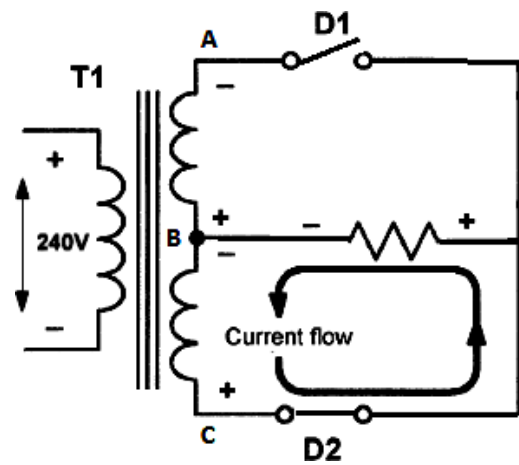


Fig.9. a) Bi-phase rectifier circuit for +ve half cycles b) Bi-phase rectifier circuit for -ve half cycles

- NOTE: i) V_{peak} produced by each of secondary windings = $17\text{V} - 0.7\text{V} = 16.3\text{V}$
 ii) Pulses of voltage developed across $R_L = 100\text{Hz}$ (if primary is 50Hz)

Bi-phase rectifier with C filter.

Two diodes D_1 and D_2 are used in this circuit. They feed a common load resistor R_L , with the help of a center tapped transformer as shown in the fig.10.

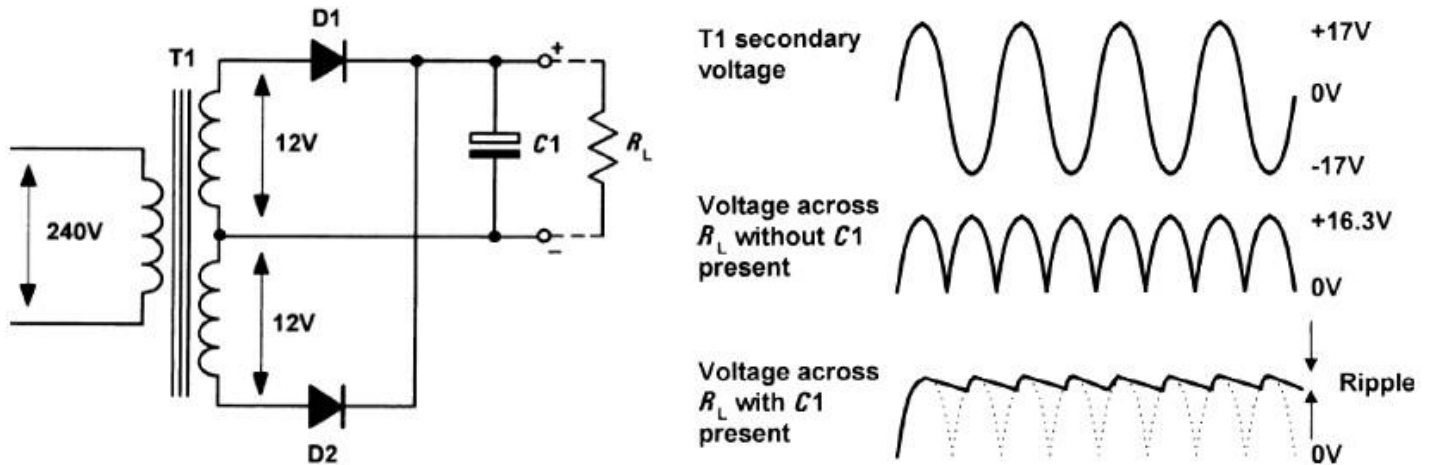


Fig.10. a) Bi-phase rectifier with C1 filter

b) Input output wave forms

When diode D_1 conduct, C_1 charges to the peak value (16.3V) of the +ve half cycle. When diode D_2 is in non-conducting state, C_1 discharges slowly through the load R_L . Similarly, when diode D_2 conduct, C_1 charges to the peak value of the -ve half cycle and C_1 starts to discharge during diode D_1 non-conducting state. Note that in this case capacitor C_1 charge and discharge twice through R_L during one full cycle.

Charging Time of C_1 to the peak value = $R_{\text{series}} \times C_1$

$C_1 R_{\text{series}} = R_{\text{secondary winding}} + R_{\text{diode}} + R_{\text{wiring and connections}}$

Hence C_1 charges quickly as soon as diode conducts.

Discharging Time of $C_1 = R_L \times C_1$

Practically, R_L is very much larger than R_{series}

Hence C_1 discharges slowly through R_L .

Disadvantages of Bi-phase Rectifier:

- It is difficult to construct and locate the center-tap on secondary winding of the transformer.
- The diodes used must have high PIV.

Bridge Rectifier Circuits.

Bridge full wave rectifier employs four diodes, but only two diodes will conduct during each half cycle.

The AC mains (240V) is applied to the primary of T1 and secondary windings providing 12V r.m.s, as shown in the fig.11.

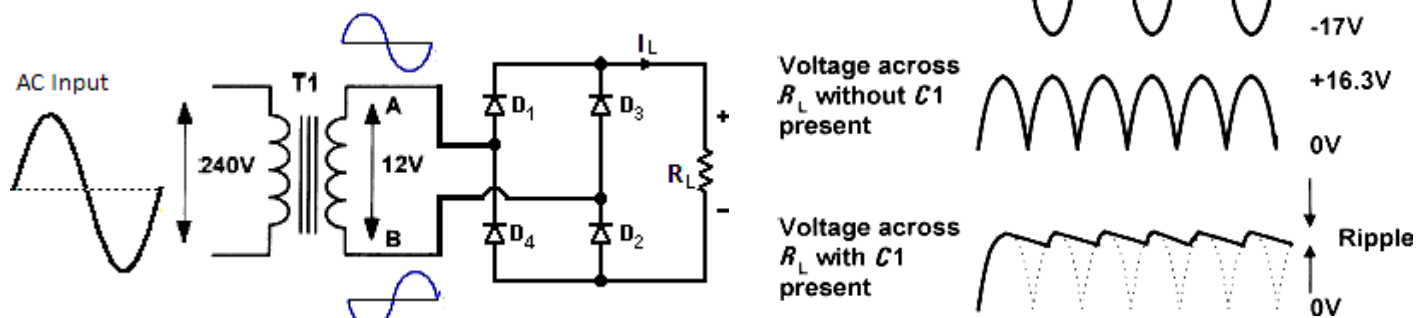


Fig.11. a) Bridge rectifier

b) Input output wave forms

During positive half cycle:

Point A will be +ve with respect to point B, then diodes D_1D_2 are forward biased act like closed switches, and hence conduct. While, diodes D_3D_4 are reverse biased act like open switches, hence do not conduct.

During negative half cycle:

Point B will be +ve with respect to point A, then diodes D_3D_4 are forward biased act like closed switches, and hence conduct. While, diodes D_1D_2 are reverse biased act like open switches, hence do not conduct.

In both +ve and -ve half cycles current I_L flow through load resistance R_L . The complete input-output voltage waveforms of the bridge full wave rectifier are shown in fig. 11(b).

Bridge rectifier with capacitor filter works very similar to that of bi-phase rectifier circuit.

Voltage Regulators

Voltage regulator is a device by which output voltage V_O , is maintained constant regardless of change in the input voltage V_{in} or load R_L . The circuit diagram of the zener diode as a simple voltage regulator is shown in the fig.12.

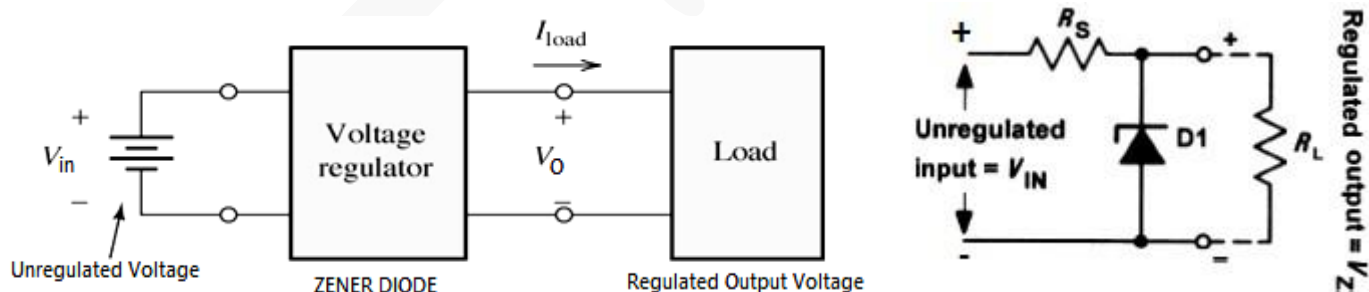


Fig.12. a) Simple block diagram of voltage regulator

b) Zener diode shunt voltage regulator

The series resistor, R_S is connected in the circuit to limit the current through the zener diode to a safe value when load R_L is disconnected. Also, the voltage drop across it is a part of unregulated input voltage, V_{in} . When R_L is connected, zener current I_Z will reduce as current ($I = I_Z + I_L$) is split into load R_L .

Output voltage V_O , remains constant until regulation fails. Regulation fails at a point at which potential divider formed by R_S and R_L produces lower voltage than V_Z voltage.

$$V_Z = V_{IN} \times \frac{R_L}{R_L + R_S}$$

Series Resistor value (ohms) = $(V_i - V_Z) / (\text{Zener current} + \text{load current})$. Maximum value of R_S can be calculated as,

$$R_{Smax} = R_L \times \left(\frac{V_{IN}}{V_Z} - 1 \right) \text{ and } R_{Smin} = \frac{(V_{IN}V_Z) - V_Z^2}{P_{Zmax}}$$

Also,

$$R_{max} = \frac{V_{i(min)} - V_Z}{I_{L(max)} + I_{Z(min)}} \text{ and } R_{min} = \frac{V_{i(max)} - V_Z}{I_{L(min)} + I_{Z(max)}}$$

The zener diode conducts the least current ($I_{Z(min)}$) when the load current I_L is maximum and it conducts the maximum current when the load current is minimum, $I = I_Z + I_L$.

The power dissipation of Zener diode is described as:

$$P_Z = V_Z I_{Z(max)}$$

Output resistance and voltage regulation

In a perfect power supply output voltage (V_O), remain constant regardless of the current taken by the load. However practically, V_O reduces as load current increases. This is due to *internal resistance* (r_i) of the power supply. That means, this internal resistance appears at the output of the power supply. It is defined as

$$R_o = \frac{\text{Change in } V_O}{\text{Change in } I_L} = \frac{dV_O}{dI_L}$$

The regulation of a power supply is given by

$$\text{regulation} = \frac{\text{Change in } V_O}{\text{Change in } V_{IN}} \times 100 \%$$

Ideally, the value of the regulation should be very small. Various regulators produce value of regulation as tabulated below:

SL.N o	Type of regulator	Regulation in %
1	Zener shunt	5 to 10
2	Sophisticated circuits based on discrete components	1 to 5
3	Integrated Circuit (IC)	Lesser than 1%

Voltage multipliers

Voltage multiplier is a modified capacitor filter circuit that delivers a dc voltage twice or more times of the peak value of the input AC voltage. Such power supplies are used for high-voltage and low-current devices such as cathode-ray tubes (the picture tubes in TV receivers, oscilloscopes and computer display).

Voltage Doubler

The circuit diagram for a full-wave voltage doubler is given in the fig.13. Assume in the beginning all capacitors are cleared (stored 0V).

During the +ve half cycle of V_{IN} voltage, diode D1 gets forward biased (conducts) and charging the capacitor C1 to a peak voltage V_{peak} with polarity indicated in the figure, while diode D2 is reverse-biased and does not conduct. During the -ve half-cycle, diode D2 being forward biased (conducts) and charges the capacitor C2 with polarity shown in the figure, while diode D1 does not conduct.

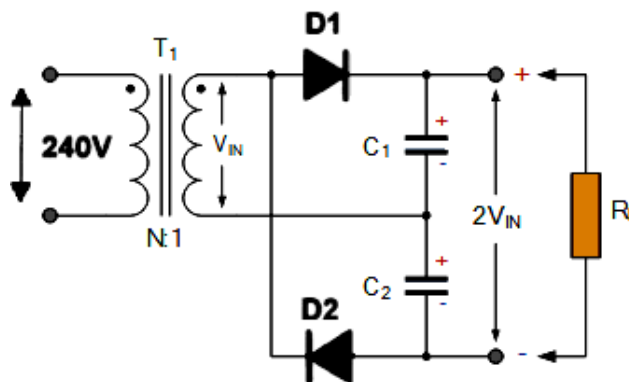


Fig.13. Voltage Doubler circuit

With no load connected to the output terminals, the output voltage will be equal to sum of voltages across capacitors C1 and C2.
i.e., $V_{C1} + V_{C2} = 2 V_{IN}$

When the load is connected to the output terminals, the output voltage V_L will be less than $2 V_{IN}$.

$$V_{out} = 2V_{IN} - \text{voltage drop across di}$$

Voltage Tripler

The voltage doubler can be extended to produce 3 times voltages (Tripler) using the cascade arrangement shown in Fig. 14. Here C1 charges to the positive secondary voltage V_{IN} , while C2 and C3 charge to twice the positive secondary voltage. The result is that the output voltage is the sum of the voltages across C1 and C3 which is three times the voltage that would be produced by a single diode.

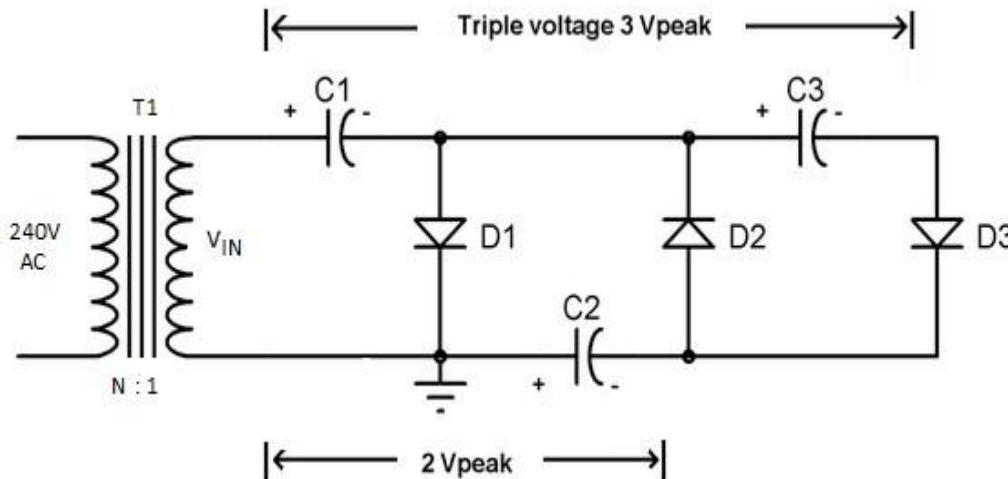


Fig.14. Voltage Tripler circuit

Assume in the beginning all capacitors are cleared (stored 0V).

During the first positive half cycle, diode D1 conducts (forward biased) and capacitor C1 gets charged the V_{IN} of secondary.

During the negative half cycle, diode D2 is forward biased and diode D1 is reverse biased. D1 does not let discharge the capacitor C1, so voltage across C1 = V_{IN} . The capacitor C2 gets charged with the combined voltage of C1 (V_{IN}) and negative peak voltage of secondary voltage, so, C2 gets charged to $2V_{IN}$.

During the second positive half cycle, diode D1 and D3 conduct and D2 get reverse biased. So, the capacitor C2 charges the capacitor C3 up to $2V_{IN}$. Now, as we can see that the capacitors C1 and C3 are in series so the total voltage across these capacitors is $V_{IN} + 2V_{IN} = 3V_{IN}$. This is how the tripled value of the applied voltage available at the output. Practically, some of the voltage drops across the diodes.

$V_{out} = 3V_{IN} - \text{voltages drop across diodes}$

Problems

1.A mains transformer having a turns ratio of 44:1 is connected to a 220 V r.m.s. mains supply. If the secondary output is applied to a half-wave rectifier determine the peak voltage that will appear across a load.

Solution

The r.m.s. secondary voltage will be given by:

$$V_S = V_P / 11 = 220 / 44 = 5 \text{ V}$$

The peak voltage developed after rectification will be given by:

$$V_{PK} = 1.414 \times 5 \text{ V} = 7.07 \text{ V}$$

2. The R - C smoothing filter in a 50 Hz mains operated half-wave rectifier circuit consists of $R_1 = 100 \text{ } \Omega$ and $C_2 = 1,000 \text{ } \mu\text{F}$. If 1 V of ripple appears at the input of the circuit, determine the amount of ripple appearing at the output.

Solution

First we must determine the reactance of the capacitor, C_1 , at the ripple frequency (50 Hz):

$$X_C = \frac{1}{2\pi fC} = \frac{1}{6.28 \times 50 \times 1,000 \times 10^{-6}} = \frac{1,000}{314} = 3.18 \text{ } \Omega$$

$$V_{\text{ripple}} = 1 \times \frac{X_C}{\sqrt{R^2 + X_C^2}} = 1 \times \frac{3.18}{\sqrt{100^2 + 3.18^2}}$$

From which:

$$V = 0.032 \text{ V} = 32 \text{ mV}$$

3. The L - C smoothing filter in a 50 Hz mains operated half-wave rectifier circuit consists of $L_1 = 10 \text{ H}$ and $C_2 = 1,000 \text{ } \mu\text{F}$. If 1 V of ripple appears at the input of the circuit, determine the amount of ripple appearing at the output.

Once again, the reactance of the capacitor, C_1 , is $3.18 \text{ } \Omega$ (see Example 6.2). The reactance of L_1 at 50 Hz can be calculated from:

$$X_L = 2\pi fL = 2 \times 3.14 \times 50 \times 10 = 3,140 \text{ } \Omega$$

The amount of ripple at the output of the circuit (i.e. appearing across C_1) will be approximately given by:

$$V = 1 \times \frac{X_C}{X_C + X_L} = 1 \times \frac{3.18}{3140 + 3.18} \approx 0.001 \text{ V}$$

Hence the ripple produced by this arrangement (with 1 V of 50 Hz a.c. superimposed on the rectified input) will be a mere 1 mV. It is worth comparing this value with that obtained from the previous example!

4. A 5V zener diode has a maximum rated power dissipation of 500 mW. If the diode is to be used in a simple regulator circuit to supply a regulated 5 V to a load having a resistance of $400 \text{ } \Omega$, determine a suitable value of series resistor for operation in conjunction with a supply of 9 V.

$$R_s \text{ max.} = R_L \times \left(\frac{V_{DS}}{V_Z} - 1 \right)$$

thus:

$$R_s \text{ max.} = 400 \times \left(\frac{9}{5} - 1 \right) = 400 \times (1.8 - 1) = 320 \text{ } \Omega$$

Now we need to determine the minimum value for the series resistor, R_s :

$$R_s \text{ min.} = \frac{V_{DS} V_Z - P_Z^2}{P_Z \text{ max.}}$$

thus:

$$R_s \text{ min.} = \frac{(9 \times 5) - 5^2}{0.5} = \frac{45 - 25}{0.5} = 40 \text{ } \Omega$$

Hence a suitable value for R_s would be $150 \text{ } \Omega$ (roughly mid-way between the two extremes).

5. The following data was obtained during a test carried out on a d.c. power supply:

(i) *Load test*

Output voltage (no-load) = 12 V

Output voltage (2 A load current) = 11.5 V

(ii) *Regulation test*

Output voltage (mains input, 220 V) = 12 V

Output voltage (mains input, 200 V) = 11.9 V

Determine (a) the equivalent output resistance of the power supply and (b) the regulation of the power supply.

The output resistance can be determined from the load test data:

$$R_{out} = \frac{\text{change in output voltage}}{\text{change in output current}} = \frac{12 - 11.5}{2 - 0} = 0.25 \, \Omega$$

The regulation can be determined from the regulation test data:

$$\text{Regulation} = \frac{\text{change in output voltage}}{\text{change in line (input) voltage}} \times 100\%$$

thus

$$\text{Regulation} = \frac{12 - 11.9}{220 - 200} \times 100\% = \frac{0.1}{20} \times 100\% = 0.5\%$$